DAY 10 · BEGINNER · DOCKER + FREE API

RAG with RAGWire

Yesterday you built the parts. Today one config file wires them together, and the assistant learns to say no.

RAGWire | citations | refused | Qdrant | Notebook

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TEN DAYS, THREE WEEKS, ONE NOTEBOOK A DAY

The 10-Day Course Map

01

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WEEK 1 TALKING TO A MODEL

01 Your First AI Program
Ask a model, print the answer

02 Roles and Messages
The system message sets the job

03 Prompt Templates
One template, many inputs

YOU WILL HAVE BUILT
a script that answers anything you type into it

WEEK 2 BUILDING SOMETHING

04 Chains: The Pipe
`prompt | model | parser`

05 Structured Output
An object, not a paragraph

06 Memory
A checkpointer and a thread id

YOU WILL HAVE BUILT
a chatbot that remembers what you said a minute ago

WEEK 3 GIVING IT ABILITIES

07 Tools
Your Python, called by the model

08 Your First Agent
It decides when to call them

09 Chunks and Vectors
Find a paragraph by meaning

YOU ARE HERE

10 RAG with RAGWire
Citations, and a refusal

One notebook and one deck a day, **60 to 90 minutes** each. Every model used is a **free** one, and Docker only turns up on Day 9.

RAG: Retrieve, Augment, Generate

1 · RETRIEVE

The chunks closest to the question

all of Day 9

→

2 · AUGMENT

Paste them into the prompt as context

Day 3, with retrieved text

→

3 · GENERATE

An answer allowed to use only those

the new part

Step 3 is where yesterday's failure gets caught,
by a sentence in a prompt, not by anything clever.

You already built steps 1 and 2. Today is about step 3.

One Config File Wires It Together


```yaml
embeddings:
provider: fastembed
model_name: BAAI/bge-small-en-v1.5

llm:
provider: openrouter
model: ...:free

vectorstore:
url: http://localhost:6333

splitter / retriever: top_k: 3
```

turns text into numbers · **Day 9**

the free model · **Day 1** · the only key needed

the Qdrant container · **Day 9**

chunking and how many to fetch · **Day 9**

nothing here is new

Four blocks of YAML, and **every one of them is a day you have already done**.

SAME RETRIEVAL FAILURE, A DIFFERENT OUTCOME

The Assistant Learns to Refuse

04

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"What is the price of a Kindle Paperwhite?"

RETRIEVAL RETURNED THREE CHUNKS ANYWAY, OUT OF 952

```
amazon-10-k-2024.pdf # consumers, sellers, developers, enterprises...
amazon-10-k-2024.pdf # unspecified amount of damages, enhanced damages...
amazon-10-k-2024.pdf # Level 2 valuations based on observable inputs...
```

a 10-K never states a product price

```
answer.refused
True
```

"The retrieved documents do not contain enough information."

Retrieval failed the same way it did yesterday. **The difference is what happened next.**

One Prompt Rule Makes It Refuse

DEFAULT_SYSTEM_PROMPT

1. Use only the sources.
2. Cite with [n] after each claim.
3. If the sources do not answer it, say so.
4. ... 5. ...

REFUSAL_SENTINEL

`"INSUFFICIENT_CONTEXT"`

the model emits this exact string, and spotting it sets **`answer.refused`**

Rule 3 is the whole day.
Without it, the model just writes a fluent paragraph.

A real boolean is what makes it usable: `if answer.refused: hand_off_to_a_human()`

What You Built in Ten Days

DAY 1 a model call	**DAY 2** the message list	**DAY 3** a reusable prompt	**DAY 4** a chain	**DAY 5** typed output
DAY 6 memory, both kinds	**DAY 7** tools	**DAY 8** an agent	**DAY 9** chunks and vectors	**DAY 10** RAG

Everything beyond this course is the same ideas at a larger size.

The one habit worth keeping:
Run the cell before you believe the sentence.

These systems fail quietly, in fluent English. Checking is the only defence.